

MICHAEL MATTIE

Senior Software Engineer — Distributed Systems, Cloud & Security
codermattie@runbox.com | bitcathedrals.github.io | Spokane, WA

SUMMARY

Software engineer with 25+ years of experience across security, distributed systems, and cloud infrastructure. Built and operated production systems at Amazon, Oracle, Comcast, and Cred.ai spanning REST/MicroService architectures, AWS Serverless platforms, and enterprise security tooling. Deep background in Python, Java, and C++ with full lifecycle experience from system design through DevOps and production operations.

CORE SKILLS

- Languages: Python, Java, C++, Perl, Common Lisp/Clojure, Shell
- Architecture: REST APIs, MicroServices, Distributed Systems, System Design, Event/Streaming Integration
- Cloud & DevOps: AWS (EC2, Lambda, API Gateway, CloudFormation, CloudWatch, S3), Serverless, Docker, CI/CD
- Security: Static Analysis Tooling, AAA (Authentication/Authorization/Auditing), Cloud Infrastructure Auditing
- Databases: SQL Server, DynamoDB, Aurora, Redshift, Postgres — schema design & normalization (UW Certificate)

PROFESSIONAL EXPERIENCE

Gauge Security — Architect, Principal Engineer (Owner) [2022-2024]

- Conceived, designed, and built a novel security auditing tool operating at enterprise scale.
- Built Python-based workflow tooling covering git/git-flow, virtualenv and package management, environment management, build packaging, and repeatable release practices, with high-quality documentation.

Cred.ai — Software Engineer [October 2020 - August 2021]

- Worked extensively with third-party APIs, addressing vendor availability and performance issues using patterns such as Retry and Circuit Breaker within a largely outsourced architecture.
- Learned and applied serverless design and AWS architecture patterns.

Amazon — Software Engineer [April 2019 - Dec 2019]

- Built a web application following 12-factor principles: a Python 3 / Flask REST API backend with an Angular 8 SPA front end.
- Wrote a backend integrating with Redshift and internal REST APIs operated by data scientists, modeling network construction activity for the Amazon networking organization.
- Built a polling/streaming SSH watcher enabling engineers to observe remote command execution in-browser.

Oracle — Software Engineer [September 2018 - March 2019]

- Built AAA (Authentication, Authorization, Auditing) intrusion detection tooling on the Oracle Cloud control plane.
- Worked on a security program designed to run throughout the Oracle Cloud; contributed to system design and scaling efforts.
- Used CI/CD tooling across the stack: virtualenv, Rundeck, Artifactory, Chef, Docker, and Odin (Oracle's Kubernetes-like orchestrator); monitored systems with Grafana.

CTS/Comcast — Software Engineer [June 2016 - August 2018]

- Built solutions at scale, integrating a distributed microservice platform with clustered video processing.
- Worked within a microservice environment of 22–24 core services (200+ total services and agents) built on a custom distributed architecture connected via REST APIs.
- Operated and configured a video streaming and metadata platform serving major media partners including AMC, using Postman and curl for daily API operations over nearly three years.

Cobalt / ADP — Software Engineer (Contract, extended)[September 2013 - December 2013]

- Delivered a 3-month contract (extended) covering Perl, ETL pipelines, database work, and performance optimization — first Seattle-area role after relocating from Spokane.

OpenEye — Software Engineer [2012]

- OpenEye: Java/JSF/PrimeFaces application development.

Security Companies & Startups — Senior Engineer / Developer

- SpyLok: Java, userspace filesystem development, remote file access, macOS.

ToolBuilders — *Senior Engineer / Developer* [June 1999-March 2003]

- Architected and developed a static source-code analysis security tool used by intelligence agencies, including source-code-in-database architecture (C/C++).

ADDITIONAL EXPERTISE

- Functional Programming: Emacs Lisp, Common Lisp, Racket (Scheme), and Clojure since 2008 — macros, closures, currying/partial application, variadic composition.
- Linux: Debian, Red Hat, Ubuntu, Gentoo, Arch, Alpine, CentOS/RHEL, FreeBSD, OpenBSD — including custom Gentoo builds and chroot-based installs; Unix systems and sockets APIs.

EDUCATION

- Certificate in Database Management — University of Washington